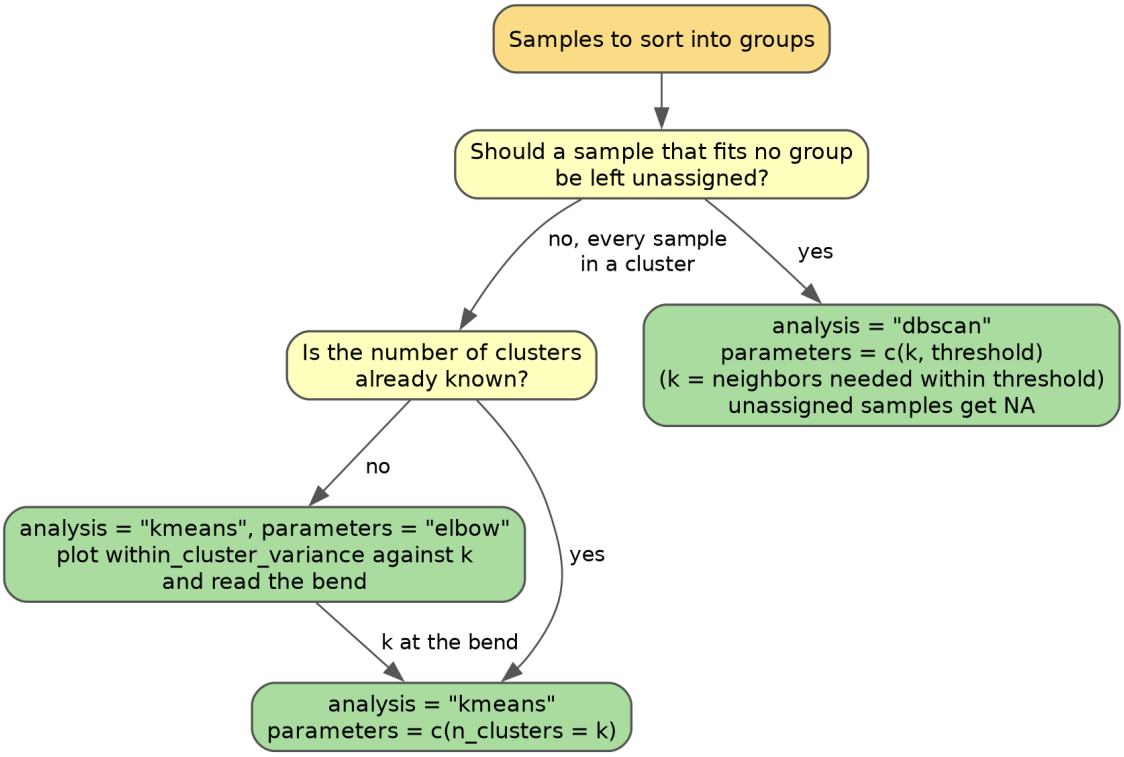
flat clustering: summary

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Flat clustering sorts samples straight into groups. **K-means** places every sample in the cluster whose center it sits closest to, so it needs the number of clusters and can never leave a sample out. **Dbscan** grows clusters outwards from points with enough close neighbors and leaves a sample that fits no cluster as `NA`. To choose the number of clusters for k-means, plot the within-cluster variance against k and read the bend (the elbow), not the lowest value: a sharp elbow means the number of groups is a property of the data, and a gentle one means it is largely a choice. Both methods scale the columns first by default. Once samples have clusters, the `cluster` column works like any other grouping column, and a grouping that also separates measurements it was never clustered on (nickel and silica, in the meteorite example) supports the idea that the groups are real.



Function or setting	What it does	What to vary
`runMatrixAnalysis(analysis = "kmeans", parameters = c(n_clusters = 3))`	sorts every sample into the number of clusters asked for	`n_clusters`
`analysis = "kmeans", parameters = "elbow"`	returns k and `within_cluster_variance` for one to ten clusters, for an elbow plot	`c("elbow", 6)` sets a different maximum
`analysis = "dbscan", parameters = c(4, 0.3)`	grows clusters from samples with at least k neighbors within a threshold distance, and leaves the rest as `NA`	k (the number of neighbors), then the threshold (measured on the scaled columns)
`columns_w_values_for_single_analyte`	names the columns to cluster on	any number of columns
`scale_variance`	scales each column before clustering (`TRUE` by default)	leave on when columns are in different units
`group_by(cluster) %>% summarize(n = n())`	counts the samples in each cluster (the `NA` row counts unassigned samples, not a cluster)	other summary statistics
`left_join(x, y, by = "fragment")`	brings back columns that were not clustered on	`by`, and `select()` the needed columns of y first
`geom_mark_ellipse(aes(fill = cluster, label = cluster))`	draws an ellipse around each cluster	use `drop_na()` on dbscan output to skip the `NA` samples